NAND flash memory

Storage and market structure in the AI era

01 Market structure

Factor	Overview
Main uses	SSDs, smartphones and USB drives. Non-volatile memory retains data without power.
Demand drivers	Growing demand for AI data-center SSDs and enterprise storage.
Price cycles	Prices tend to rise when supply is tight and fall during oversupply.
Technology competition	More layers in 3D NAND, improved yields and lower manufacturing costs.

02 Key market indicators

Quarter-on-quarter changes in NAND contract prices and inventory levels.
The sales mix and shipped capacity of data-center SSDs.
Manufacturers’ capital-spending plans and factory utilization.
Operating cash flow, capital expenditure and depreciation.

Four related Japanese companies

Business areas and exposure to the NAND market

03 Business characteristics

Tokyo Electron / 8035 - Semiconductor production equipment. Exposed to chipmakers’ investment cycles across NAND and other segments.

SCREEN Holdings / 7735 - Cleaning equipment. Supports increasingly sophisticated chip production; affected by changes in manufacturers’ investment and equipment demand.

DISCO / 6146 - Precision processing equipment. Strengths in wafer cutting, grinding and polishing. Earnings do not move directly in line with NAND prices.

Kioxia Holdings / 285A - NAND flash manufacturing. Selling prices, shipment volumes and production costs directly affect profitability.

04 Perspectives for evaluating companies

Approach	Focus
Graham	10-year average earnings, current ratio and margin of safety
Buffett	Owner earnings, competitive advantage and capital efficiency
Munger	The relationship between business quality and price